DENGHAN XU

Robotics MSc Student | Mechanical Design and Engineering
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Portfolio: https://hsuhans6-sys.github.io/
GitHub: https://github.com/hsuhans6-sys

PROFILE

Robotics MSc student at the University of Manchester with a background in mechanical design, manufacturing and automation. Practical experience in CAD modelling, mechanical assembly and engineering drawings through university projects, robot competitions and industry placements. Interested in robotic manipulation, assistive devices and combining mechanical design with sensing and control.

EDUCATION

University of Manchester, United Kingdom
MSc Robotics | Currently studying

Shaanxi University of Science and Technology, China
Ulster College | Mechanical Design, Manufacturing and Automation
Chinese–British dual-degree programme. Ranked in the top 10% of the programme. Relevant coursework included Theory of Machines, Automated Mechanical Design, Mechanics of Materials, Theoretical Mechanics and Industrial Robotics.

ENGINEERING PROJECTS

Arm Rehabilitation Assistive Device | April–June 2025
Designed the mechanical structure of an affordable device intended to support rehabilitation exercises for stroke patients. Used Solid Edge for structural design and KeyShot for exploded-view renderings. Independently conducted market research. Project materials included an English introduction and a demonstration animation.

Planar Indexing Cam Mechanism | May–July 2025
Designed a mechanism using a cam made from grade 45 steel. Generated motion simulations in UG using cam-profile coordinate data and used MATLAB to simulate the cam profile. Modelled components and assemblies in SolidWorks, including the housing and input shaft. Prepared assembly and exploded drawings while considering manufacturing processes and standard component selection.

ROBOTAC Robot Competition | June 2025
Participated in the 24th national university robot competition as part of a team awarded a national third prize. Contributed to mechanical design, physical robot assembly, and circuit and program design.

ENGINEERING EXPERIENCE

Dongfeng Special Equipment Co., Ltd. | Mechanical Engineering Intern
Shiyan, Hubei, China | July–September 2024
Supported the design and modification of non-standard mechanical equipment, including components, structures and assemblies. Used SolidWorks to create models in response to customer requirements. Prepared engineering drawings for manufacturing and supported production scheduling and delivery coordination.

Northwest Machine Co., Ltd. | Production Design Intern
Baoji, Shaanxi, China | June–July 2024
Learned about engineering design workflows, manufacturing processes and machining equipment. Gained exposure to mechanical assembly, including bearings, and developed an understanding of medical equipment and its manufacturing precision requirements.

TECHNICAL SKILLS

Mechanical design: SolidWorks, Solid Edge, CAD, CAXA, UG/NX, engineering drawings and mechanical assembly.
Visualisation: KeyShot rendering and exploded-view presentation.
Programming and robotics: MATLAB project experience; foundational Python and C++ knowledge; currently developing ROS 2 skills.
Documentation: Microsoft Office, WPS Office and technical report preparation.

LEADERSHIP

Head of Organisation Department, Ulster College
Shaanxi University of Science and Technology | September 2023–June 2024
Coordinated campus activities, including sports events and English-language activities. Participated in organising collaborative activities involving Ulster University.

Class Youth League Secretary, UC223
Shaanxi University of Science and Technology
Organised class activities and coordinated participation in themed student events.

AWARDS AND QUALIFICATIONS

National Third Prize, ROBOTAC Robot Competition, 2025.
University-level Outstanding Youth League Cadre, 2022 and 2025.
College English Test: CET-4 and CET-6.